

Python Basics Assignment - Answers

Q1. What is Python and why is it widely used in Data Analytics? Mention any three reasons.

Python is a high-level programming language. It is widely used in Data Analytics because: (1) It is easy to learn and read, (2) It has powerful libraries such as Pandas, NumPy, and Matplotlib, and (3) It supports data cleaning, analysis, visualization, and automation.

Q2. Explain the difference between List and Tuple in Python.

List: Mutable (can be modified) and uses []. Tuple: Immutable (cannot be modified) and uses ().

Q3. What is a function in Python? Why are functions useful?

A function is a reusable block of code that performs a specific task. Functions reduce repetition, improve readability, and make programs easier to maintain.

Q4. Write a Python program to take a user's name as input and print a greeting message.

```
name = input("Enter your name: ")
print("Hello,", name, "! Welcome!")
```

Q5. Write a Python program to check whether a number is even or odd using conditional statements.

```
num = int(input("Enter a number: "))
if num % 2 == 0:
    print("Even")
else:
    print("Odd")
```

Q6. Write a program to print numbers from 1 to 10 using a loop.

```
for i in range(1, 11):
    print(i)
```

Q7. Create a list of five numbers and print the maximum number from the list.

```
numbers = [10, 25, 7, 45, 18]
print(max(numbers))
```

Q8. Write a Python program to remove duplicate values from a list using a set.

```
numbers = [1, 2, 2, 3, 4, 4, 5]
unique_numbers = list(set(numbers))
print(unique_numbers)
```

Q9. Write a function that returns the square of a number.

```
def square(n):  
    return n * n  
  
print(square(5))
```

Q10. Write a Python program to count how many times a number appears in a list.

```
numbers = [2,3,4,2,5,2]  
target = 2  
print(numbers.count(target))
```